The challenge of serodiagnosis of sleeping sickness in the context of elimination

Gambiense human African trypanosomiasis (HAT), or sleeping sickness, is a neglected tropical disease that has been targeted for elimination as a public health problem by 2020.[1] Although this goal is challenging, it now seems achievable. Control efforts, sustained mainly by mobile teams who screen populations at risk with the Card Agglutination Test for Trypanosomiasis (CATT) and treat patients if confirmed by microscopic parasitological tests, have led to a rapid decrease in disease prevalence. Incidence passed below the symbolic number of 10 000 newly reported cases in 2009 and reached 7216 in 2012.[2] Furthermore, important innovations in terms of diagnosis and treatment are now in development, generating great hopes that sustainable control strategies could be implemented. Among these achievements is the forthcoming availability of lateral flow rapid diagnostic tests that might be used for serodiagnosis of infection in place of CATT, which requires an electrical supply, a cold chain, and trained personnel. Alongside the elimination process, decreased disease prevalence means that screening by mobile teams will become less and less cost effective. Thus, integration of HAT diagnosis into the endemic country health systems needs to be promoted. Rapid diagnostic tests that do not require any specific equipment and are easy to use should help greatly in the development of an effective passive surveillance system in primary health-care centres.

In *The Lancet Global Health*, Philippe Büscher and colleagues[3] present the first formal assessment under field conditions of the HAT Sero-*K*-SeT rapid diagnostic test and compared its diagnostic accuracy with two other current serological tests: CATT and immune trypanolysis (a very sensitive and specific test that requires laboratory conditions). The study was done in the Bandundu province of DR Congo during an active screening campaign. The investigators noted no significant difference in sensitivity and specificity between the three tests and concluded that the accuracy of HAT Sero-*K*-SeT is adequate for HAT serodiagnosis, and thus that the test could be implemented for passive surveillance at peripheral health facilities.

Notably, another rapid diagnostic test developed by the Foundation for Innovative Diagnostics (FIND) in collaboration with SD Bioline[4] has also been developed. The two tests are based on the antibody recognition of the same trypanosome variant surface glycoproteins (LiTat 1.3 and LiTat 1.5). Additional efforts are now needed to assess their accuracy and implementation as passive surveillance techniques in different areas in central and west Africa because the variable antigen type recognition pattern of *Trypanosoma brucei (T b) gambiense* is known to differ geographically[5–7] and because integration of HAT diagnosis in the primary health-care system is challenging.[8]

Nevertheless, availability of such rapid serodiagnostic tests only partly simplifies the diagnosis of HAT. As for CATT, these rapid diagnostic tests are only serological suspicion tests and parasitological confirmation by central laboratories is needed before treatment can be started. This issue arises because all available serological tests are hindered by insufficient specificity, and fears surround overtreatment with present drugs that are toxic, complex to administer, and require long-term admission to specific treatment centres. In this regard, new oral drugs such as fexinidazole[2,9] have now entered phase 2 clinical trials and might soon become available. Second-generation rapid diagnostic tests, based on other recombinant proteins or peptides, are also under development as continuous efforts to improve HAT diagnosis. These efforts should be encouraged because they might (alone or in combination) significantly increase the specificity of serological diagnosis and thus simplify treatment decisions. The possibility of being able to treat patients with an oral drug on the basis of a rapid diagnostic test or combination of rapid diagnostic tests would be a giant leap towards the elimination of sleeping sickness: a dream that is now in sight.

A healthy competition exists between several groups to develop HAT rapid diagnostic tests, which should improve quality and provide options in case one test fails to improve diagnostic accuracy. As a neglected tropical disease and with decreasing disease prevalence, progressive commercial disinterest is a

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possibility for HAT. Thus, reliance on a single rapid diagnostic test for which production could be stopped for economic reasons is dangerous, and could harm the elimination strategy. All the political, financial, and technical ingredients seem to be in place to make HAT elimination a success story.[10] We should not miss this historical opportunity.

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We declare that we have no competing interests.

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